

Dominicus Johan Nararya

M: 022 357 1401 | E: jojonararya@gmail.com

Education

The University of Auckland

Master of Science Majoring in Data Science

- Courses: Applied Multivariate Analysis, Advanced Regression, Machine Learning, Econometrics, Advanced Design & Analysis of Algorithms, Big Data Management.

Bachelor of Science Majoring in Data Science

- Courses: Statistical Computing, Statistical Modelling, Statistical Learning, Statistical Theory, Optimisation and Data-driven Decision Making, Artificial Intelligence, Data Structures & Algorithms.

Technical Skills

Programming Languages: Python, R, SQL, SAS.

Libraries/Frameworks: Pandas, Numpy, Polars, Sci-kit Learn.

Projects

Anomaly-detection in Prediction Markets (ongoing)

Jul 2025 Present

- Developed a robust ETL pipeline in Python to extract, transform, and load raw trading data, processing over transactions into a clean, structured CSV for analysis.
- Engineered features to profile trader behaviour, including Return on Investment (ROI), trade timing relative to key events, and win/loss ratio.
- Currently Implementing and evaluating multiple unsupervised machine learning models, including **Logistic Regression** and **Local Outlier Factor (LOF)**, to identify traders exhibiting anomalous profitable patterns.

Researching HRL in Atari Gymnasium Environments

Jul 2025 Nov 2026

- Formulated the primary research question and conducted a comprehensive literature review to establish the project's theoretical foundation and scope for the proposal presentation.
- Engineered a custom reward wrapper for the reinforcement learning environment in the Atari game *Montezuma's Revenge* to facilitate effective agent training.
- Executed a rigorous performance evaluation of models using the rliable library, providing robust statistical analysis and benchmarking of experimental outcomes.

Brief Assessment of Multi-variate techniques in detecting market regime shifts

Feb 2025 June 2025

- Engineered an end-to-end data pipeline to process high-frequency, 2-minute financial data for 480 S&P 500 constituents, preparing the dataset for analysis by calculating and standardizing logarithmic returns.
- Implemented a dynamic analysis model using rolling window Principal Component Analysis (PCA) over a 5-day window to capture the time-varying structure of the market, using the resulting component loadings as features for K-means clustering.
- Successfully identified two possible distinct market regimes, validating the clusters by overlaying SPY and VXX returns to characterize them as a low-volatility "risk-on" state and a high-volatility "risk-off" state

Experience

Reservist (Part-time)

Nov 2021 Present

New Zealand Defence Force

- Trained and coached others in the effective use of weapon systems, with strong attention to detail and adherence to instructions.
- Adapted quickly to dynamic environments, maintaining performance and focus in challenging situations

Activities

Volunteer

Aug 2020 Jan 2023

Hospice New Zealand

- Volunteered as a cashier at a hospice charity shop, gaining valuable experience in customer service while supporting a worthy cause.